



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 28.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BITS F364
Course Title : Human Computer Interaction
Instructor-in-Charge : Prof. Siddharth Mehrotra (siddharth.mehrotra@pilani.bits-pilani.ac.in)
Instructor(s) :
Tutorial/Practical Instructors:

1. Course Description: Introduction to HCI in the AI era; Understanding People: Perception, Motor control, Cognition, Needs, and Experience; User Research: Interviews, Field research, and Survey research; Understanding Interaction: Usability Engineering, Information and control, Dialogue, Tool use, Automation, and Rationality; User Interfaces: Input devices, Displays, Interaction techniques, and Reality-based interaction; Design: Cognition, Practice, and Processes; Engineering: Systems, Design engineering, Safety and risk, and Computational representations and models; Evaluation: Analytical methods, Think-aloud studies, and Experiment.

2. Scope and Objective of the Course: At the completion of this course, students will be able to:

- **Understand** the concepts and theories of human-AI interaction. Furthermore, they will be able to explain basic concepts from the relevant theories in cognitive and social sciences. (LO1)
- **Illustrate and compare** methods of “user-centered engineering (UCE)”, involving future users in various stages of the design and development process of AI systems. (LO2)
- **Apply** “user-centered” methods appropriately, as a team; as part of design, development, and evaluation of AI systems. (LO3)
- **Apply** ethics procedures for user studies (e.g., privacy, consent) and FATE principles in design of prototype based AI systems. (LO4)
- **Describe** the perspectives and values of multiple stakeholders at different stages of the user-centered engineering process (requirements, design, and evaluation). (LO5)

3. Textbook (T1): Hornbæk, Kasper, Per Ola Kristensson, and Antti Oulasvirta, Introduction to Human-Computer Interaction (HCI) (Oxford, 2025; online edn, Oxford Academic, 21 Aug. 2025), <https://doi.org/10.1093/oso/9780192864543.001.0001> - Open Access)

4. Reference Book (R1): Human – Centered AI (HCAI) by Ben Shneiderman, 2022 (Available via BITS Library) & Research Papers (RP)

5. Course Plan:

Lecture#	Learning Objectives	Topics	Learning Outcome	Reference to Text
01-04	Understanding People	Introduction to HCI; perception, motor control, cognition, needs and motivations, experience, collaboration, communication	Explain how human perception and cognition shape interaction design	T1. Ch.1-9
05-06	User Research	Introduction to user research, interviews, field research, survey research, unobtrusive research, representations of user research	Plan and conduct user research using multiple methods	T1. Ch.10-15



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Lecture#	Learning Objectives	Topics	Learning Outcome	Reference to Text
07-08	Understanding Interaction	Information and control, dialogue, tool use, automation, rationality, practice	Model interaction as dialogue, tool use, and automation	T1. Ch.16-22
09-12	User Interfaces	Input devices, displays, interaction techniques, commands and navigation, graphical user interfaces, reality-based interaction	Compare interface technologies and interaction techniques	T1. Ch.23-29
13-16	Design	Design cognition, design practice, design processes	Apply structured design processes and cognition to practice	T1. Ch.30-33
17-21	Engineering	Systems, design engineering, safety and risk, software, computational representations and models	Connect engineering practice to safe, reliable systems	T1. Ch.34-39
22 - 26	Evaluation	Analytical evaluation methods, think-aloud studies, experiments, field evaluations	Evaluate interfaces using analytical and experimental methods	T1. Ch.40-44
27-28	Foundations of Human-AI Interaction	Introduction to human-AI interaction, practical machine learning, history of AI & HCI, guidelines for human-AI interaction	Apply guidelines and ML foundations to human-AI interfaces	R1 / RP
29-30	Engineering: Systems, Design engineering, Safety and risk I	AI alignment, AI in social decision-making, stakeholder-driven and value-sensitive design, fairness in ML and in the field, auditing, responsible AI in practice	Design and audit AI systems for fairness and accountability	R1
31-32	Engineering: Systems, Design engineering, Safety and risk II	Transparency and interpretability, usable privacy and security, speculative critique of AI futures (Black Mirror Writers' Room)	Explain transparency, privacy, and speculative AI risks	R1
3-34	Advanced Topics	Deep learning, CNNs, generative models & RL, large language models and HCI, prototyping with advanced AI, writing in the age of LLMs, peer feedback on projects	Prototype and write using advanced AI and LLM tools	RP / Project
35-38	Advanced Topics	AI agents, AI literacy, AI for innovation and learning, envisioning future (Writers' Room II)	Assess AI agents, literacy, and future interaction scenarios	RP
39-40	Final Presentations	Presentation and defense of final project outcomes	Present and defend final project work	Project

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30%	TBA	Open Book
Comprehensive Examination	3 h	40%	13/12/2026, 9 am – 12 noon	Closed Book
Group Report/Article	12 weeks	30*%	2 weeks before Compre.	Research Article

*Bonus marks for in-class participation in discussions and quizzes related to group report

7. Chamber Consultation Hour: Wednesday, 10:00 – 12:00 *by appointment*

8. Notices: All components of the evaluation scheme are mandatory to pass the course.

9. Make-up Policy:

10. Attendance Policy: Not mandatory but strongly recommended.

11. Minimum passing Marks criteria:

12. Note (if any): Group Assignments are expected to be executed in groups of minimum 2 and maximum 4 members. IC encourages the creation of teams that are balanced in terms of discipline and skills.

Instructor-in-charge
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